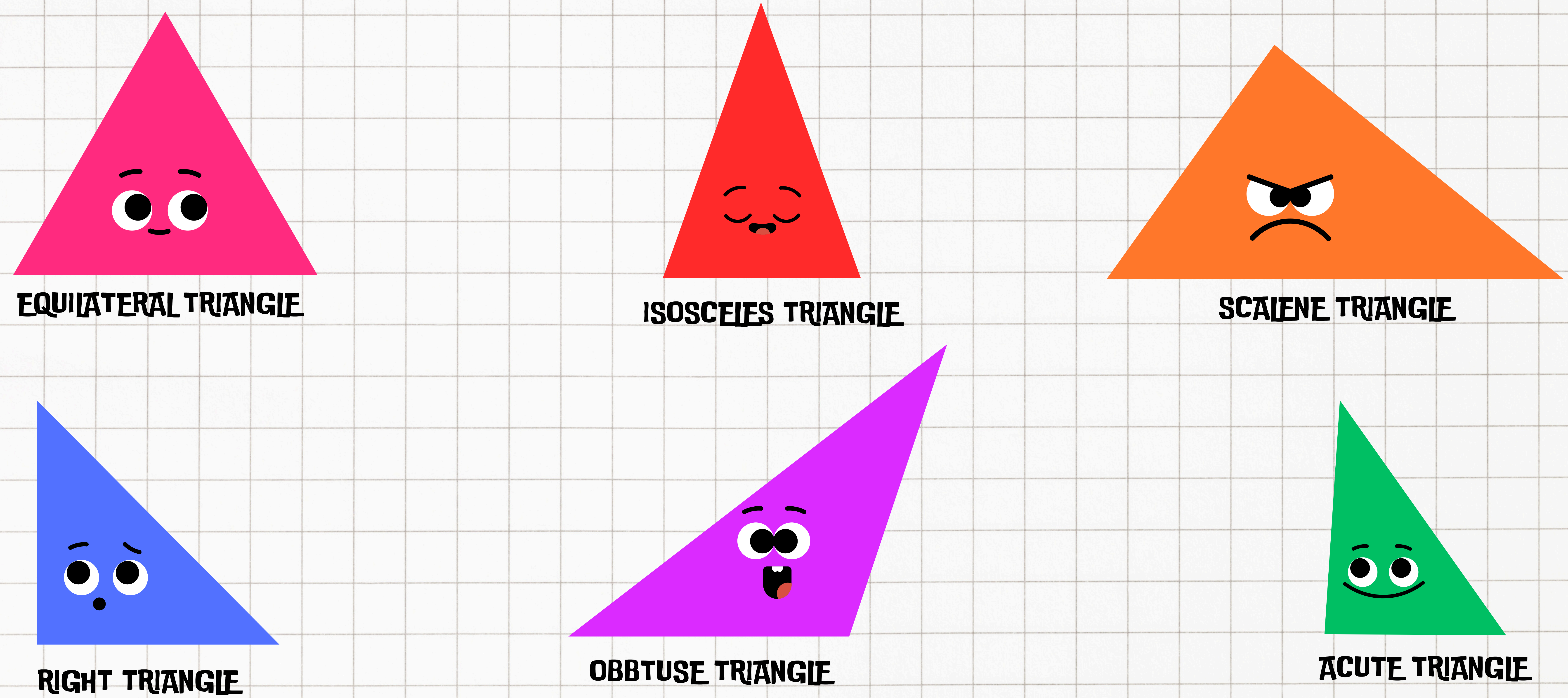
# GEOMETRY

Understanding Shapes, Angles, and Dimensions

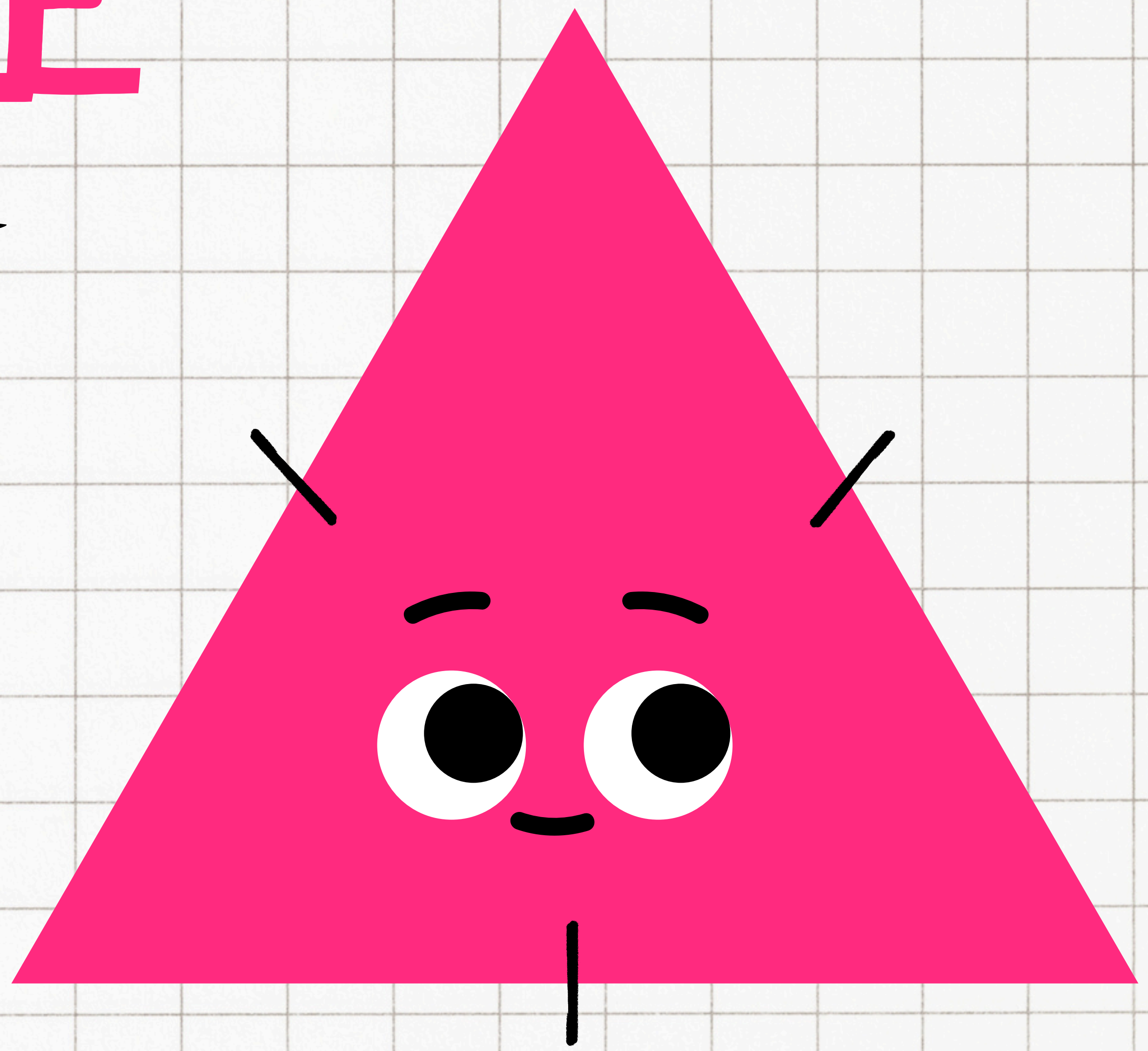
# MEET THE TRIANGLES



# EQUILATERAL TRIANGLE

<u>Definition:</u> All three sides are of equal length.

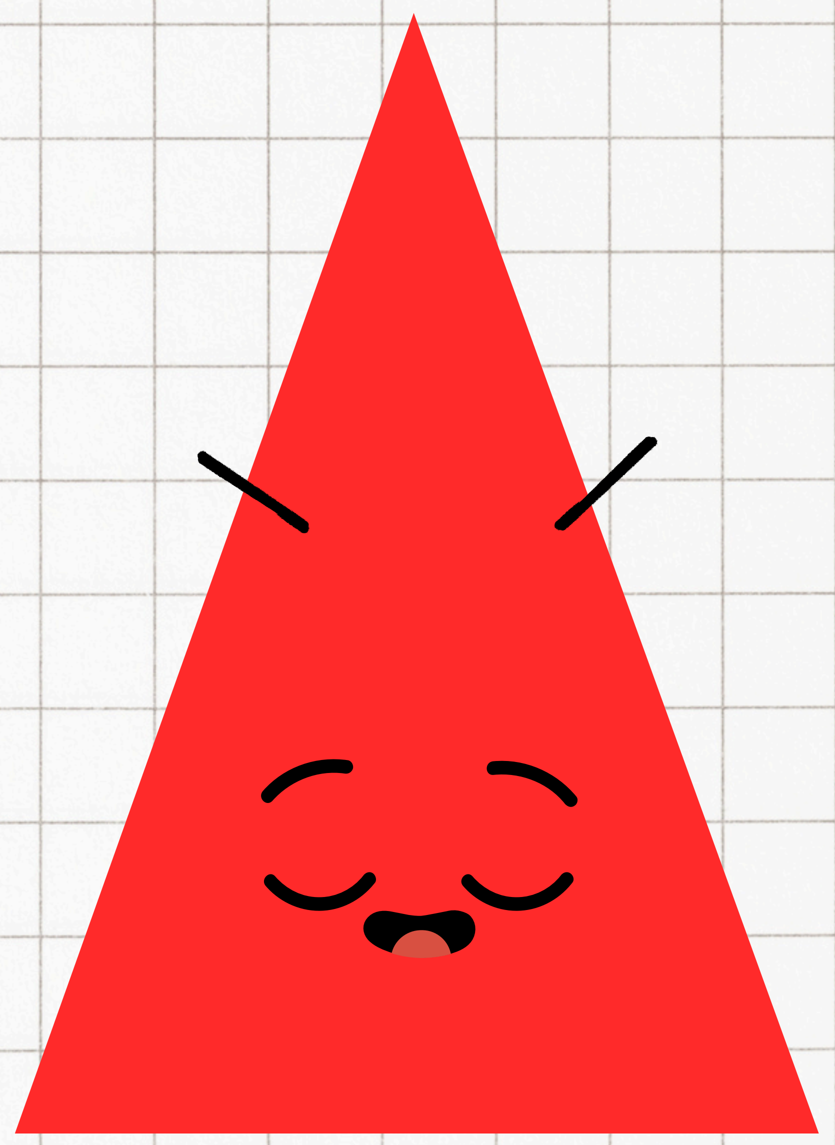
<u>Properties:</u> All three angles are equal, and each angle measures 60°.



# ISOSCELES TRIANGLE

Definition: Two sides are of equal length.

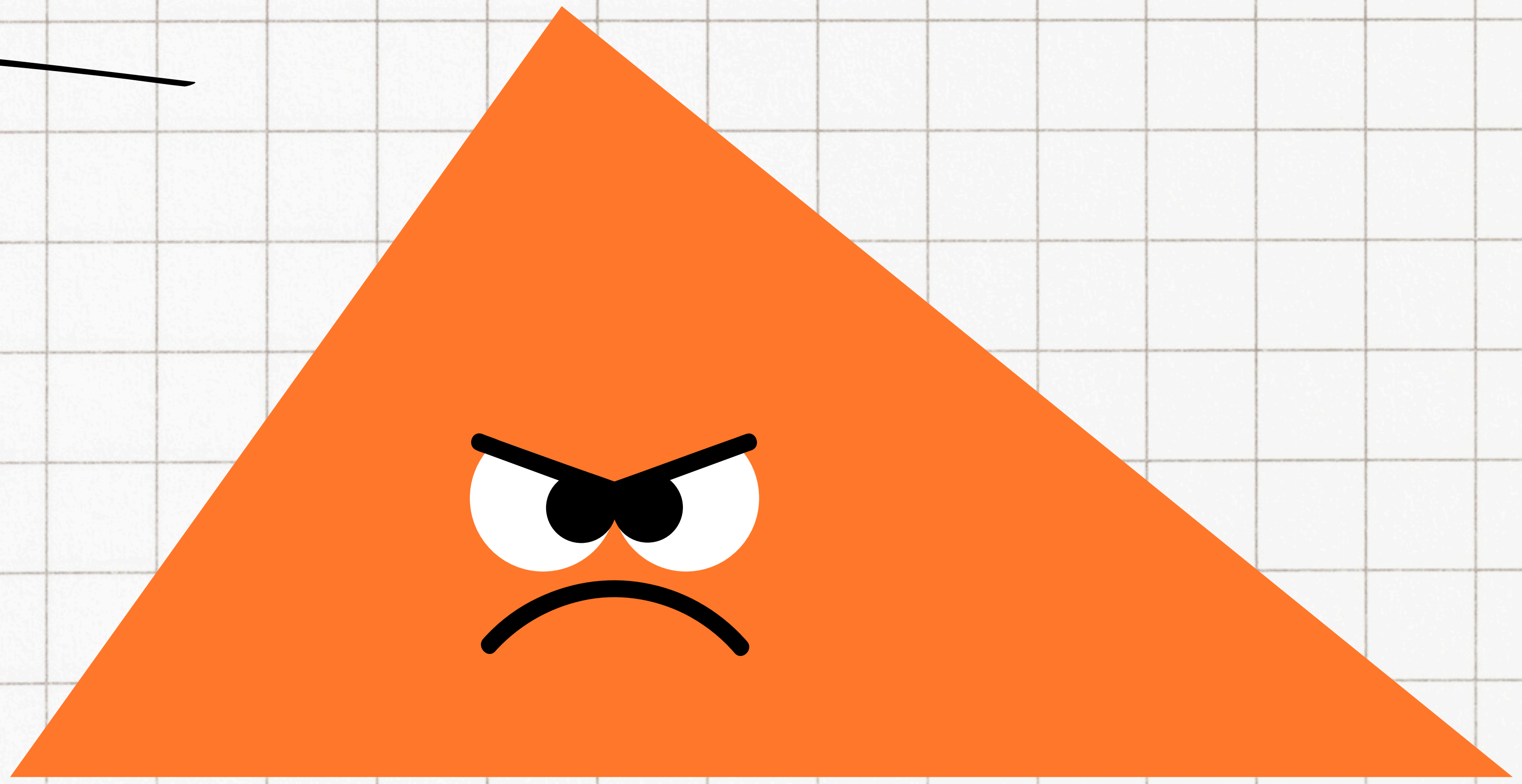
Properties: The angles opposite the equal sides are also equal.



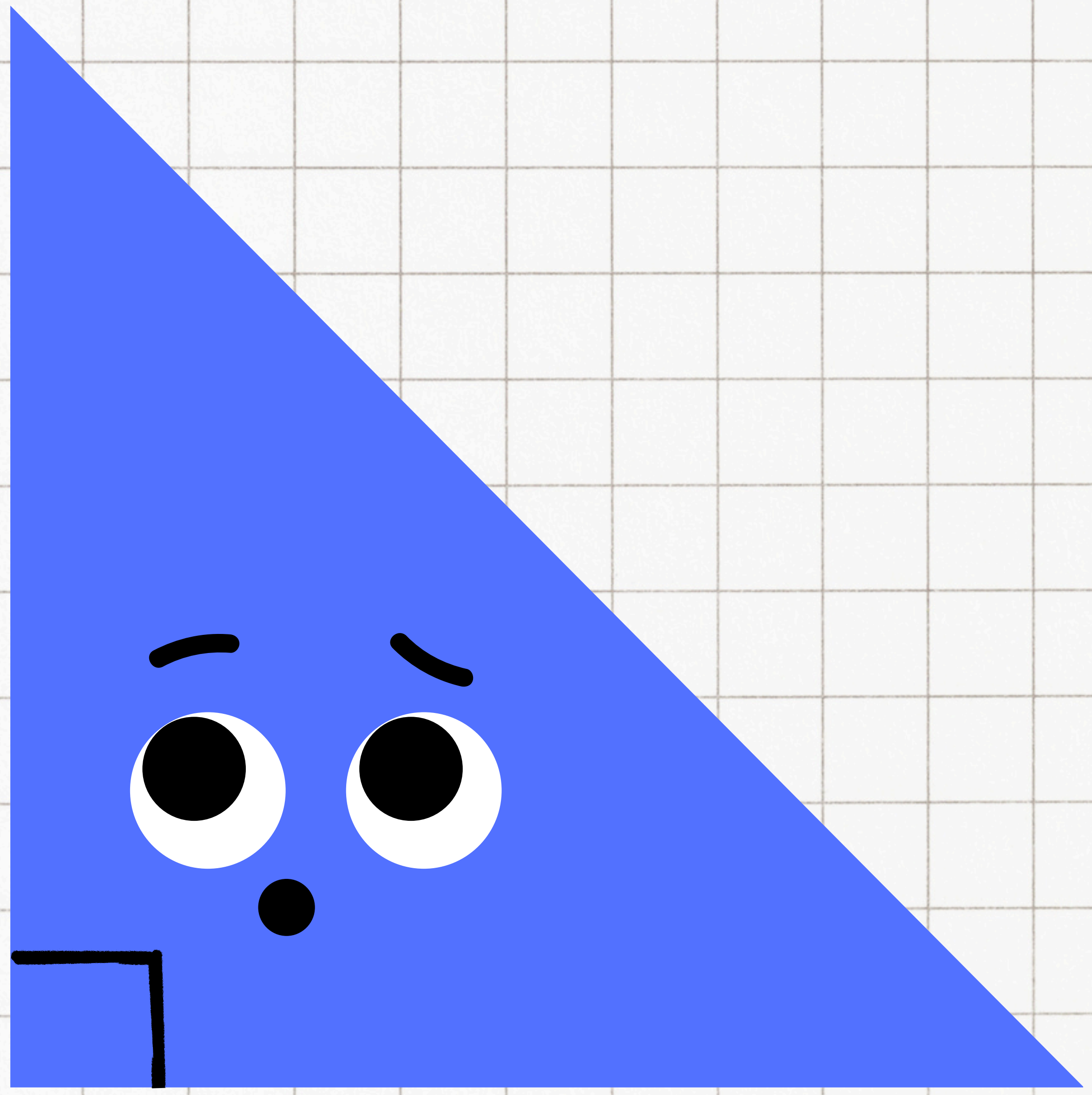
# SCALENE TRIANGLE

<u>Definition:</u> All three sides are of different lengths.

<u>Properties:</u> All three angles are different.



# RIGHT TRIANGLE



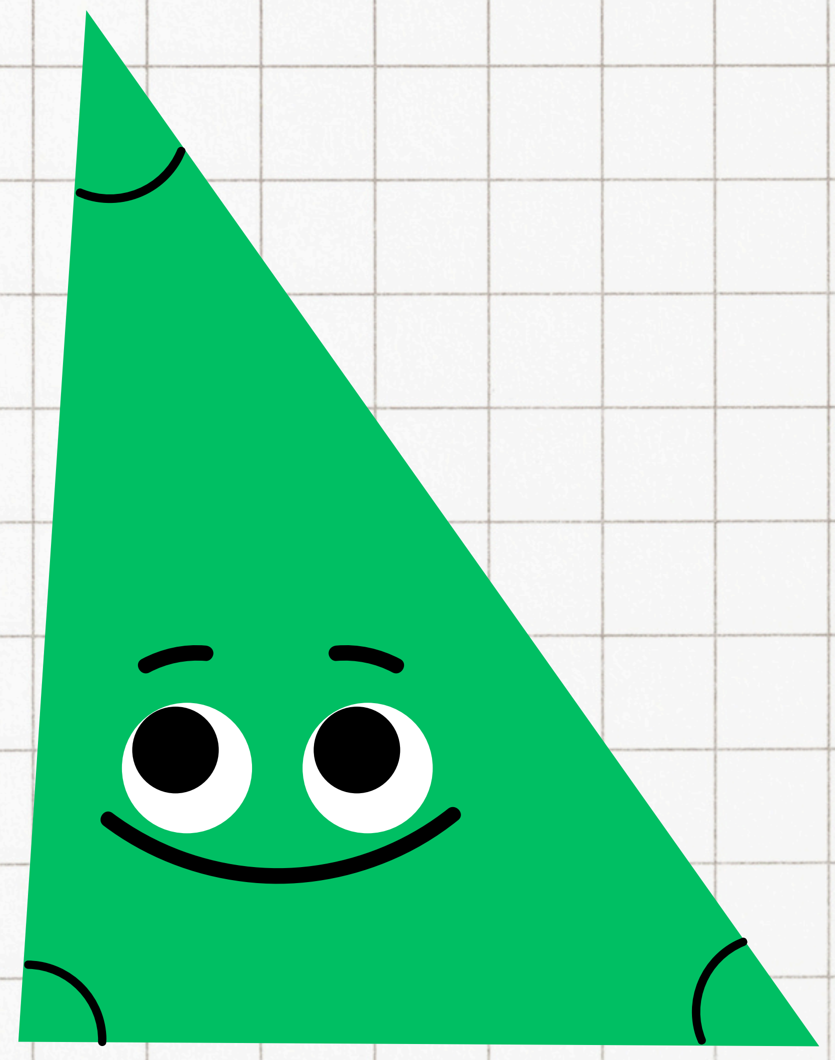
**Definition:** One of the angles is exactly 90°.

**Properties:** The side opposite the right angle is the hypotenuse, the longest side of the triangle.

# ACUTE TRIANGLE

**Definition:** All three angles are less than 90°.

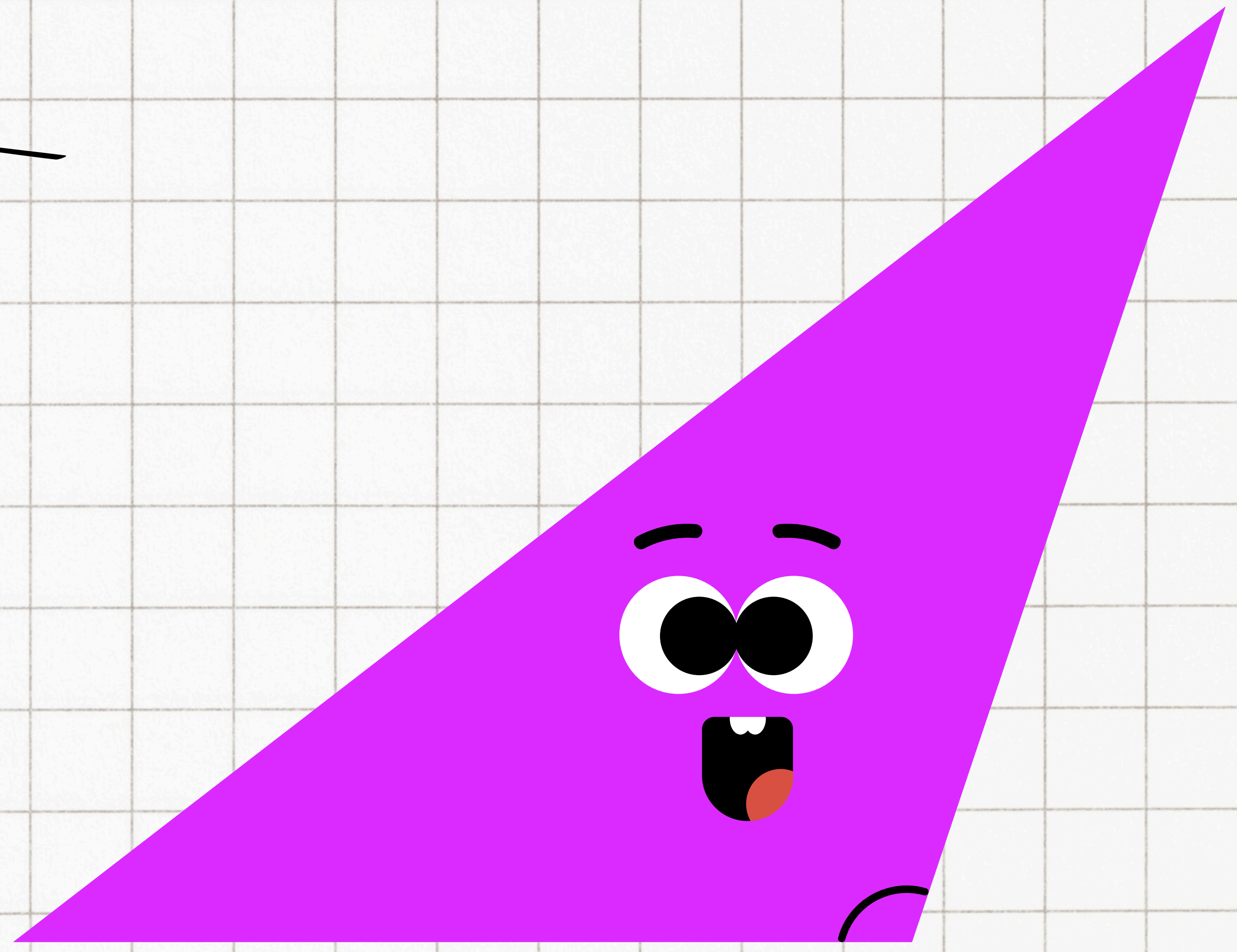
**Properties:** The triangle is sharp-angled, and the sum of the angles is still 180°.

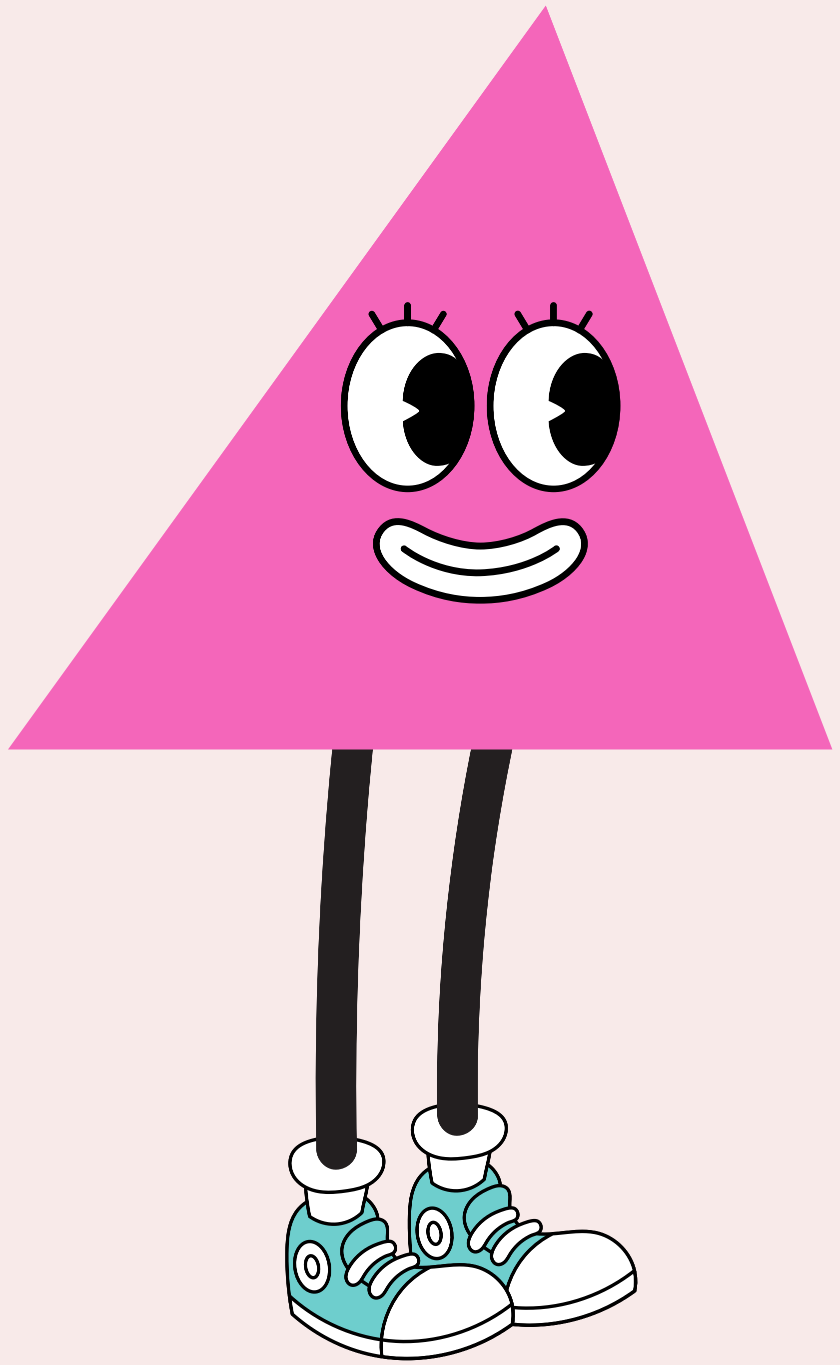


# OBTUSE TRIANGLE

<u>Definition:</u> One of the angles is greater than 90°.

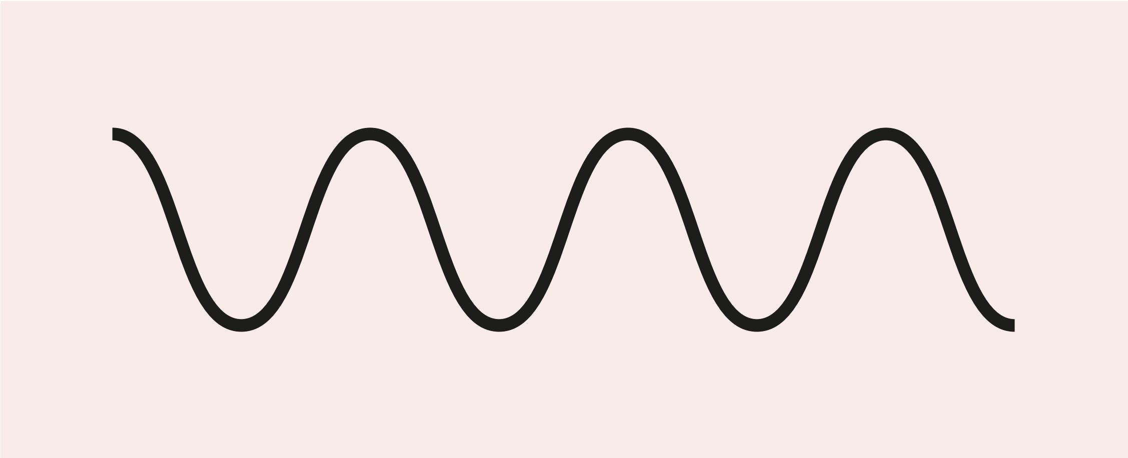
<u>Properties:</u> The other two angles are acute (less than 90°).



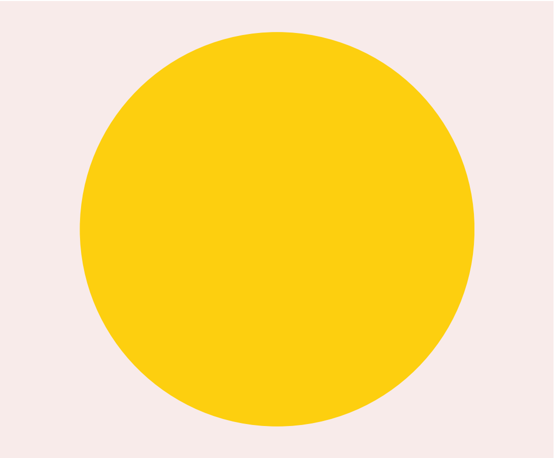


# WHAT IS A POLYGON?

- A polygon is a geometric figure made up of straight sides.
- There are no curved lines or open shapes.

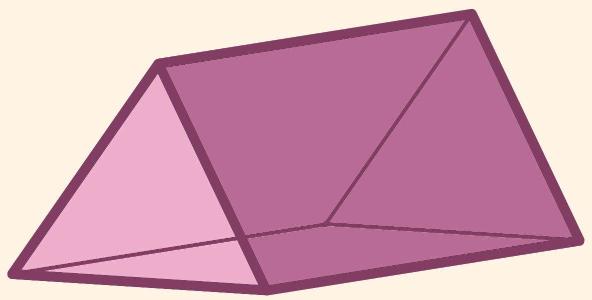
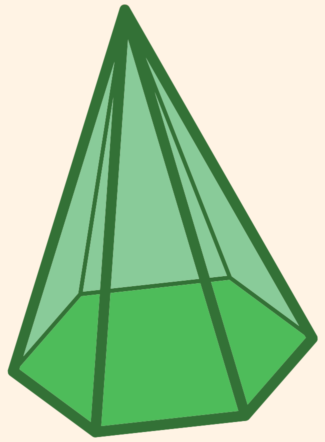
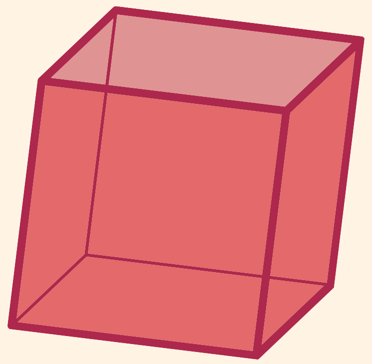
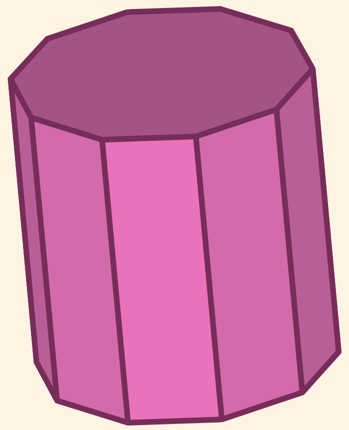
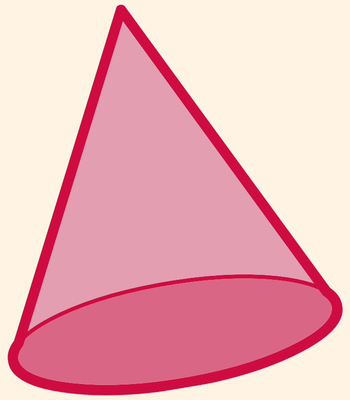


*wavy line*

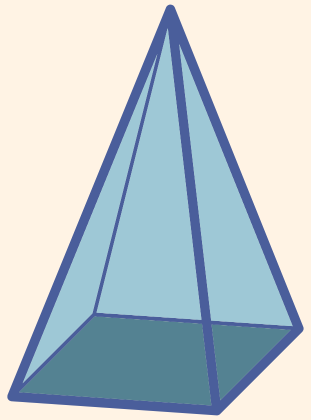
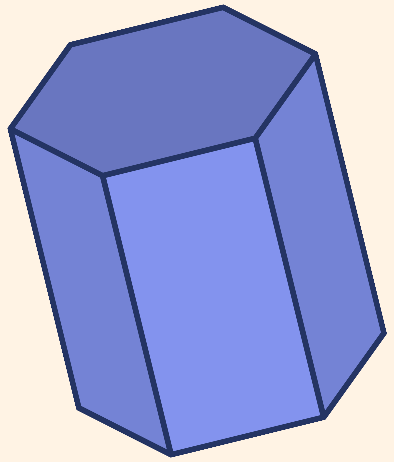
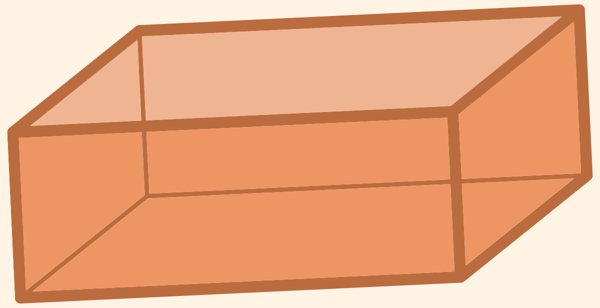
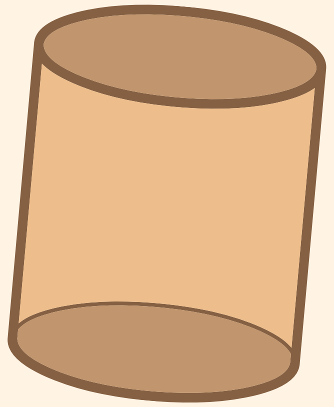
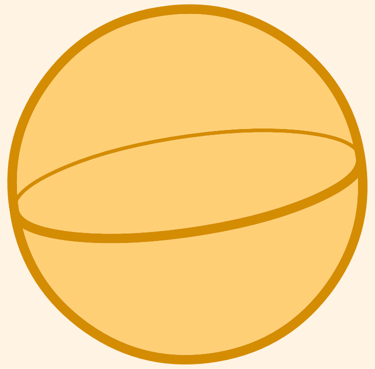


*circle*

**Are there any polygons in the pictures above?**

# 3D Shapes

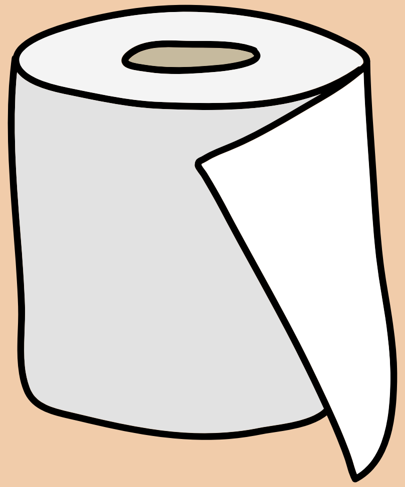






# What are 3D shapes?

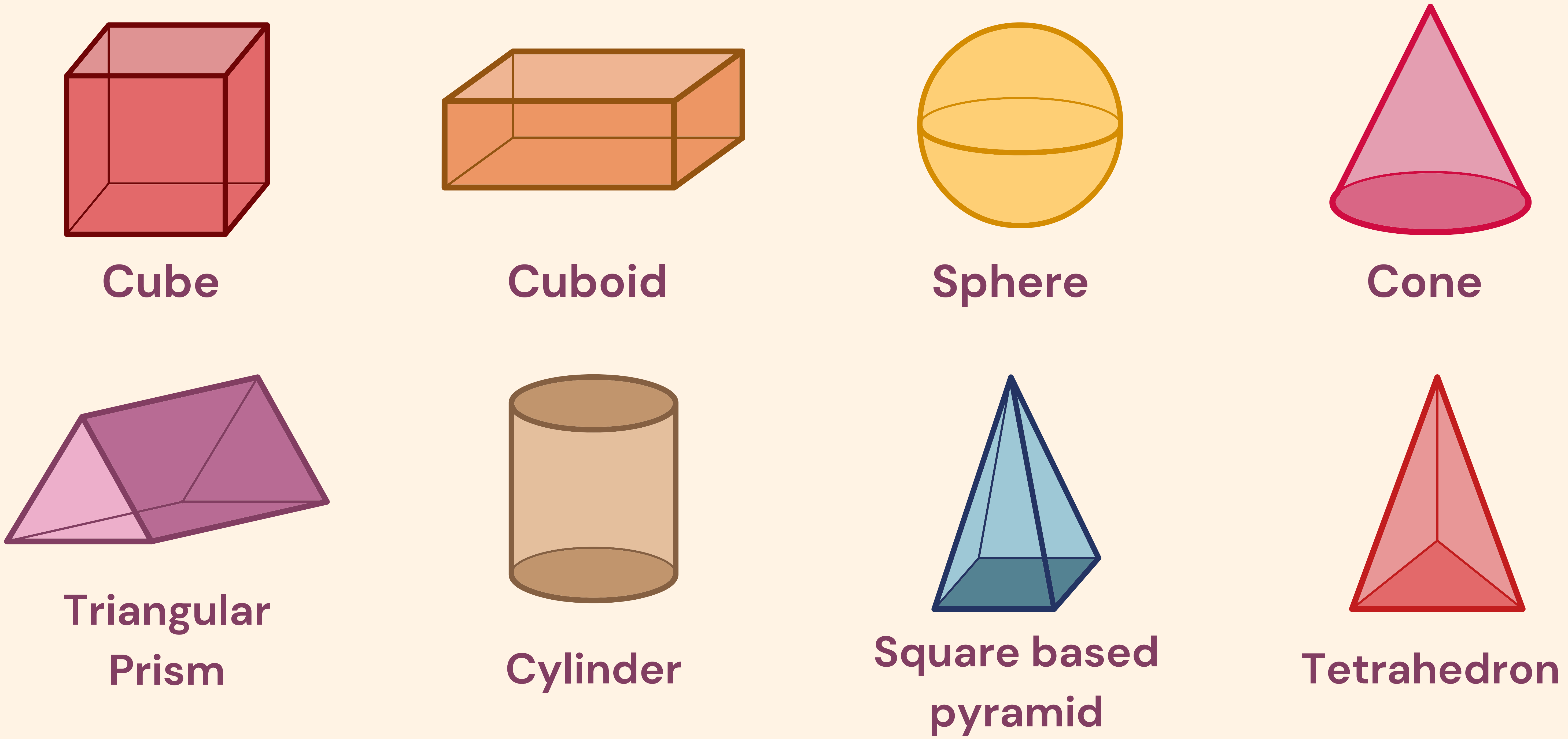
3D shapes, or three-dimensional shapes, are objects that have length, width, and height.

We will explore the mathematical names of 3D shapes and their properties.

## Real life objects:

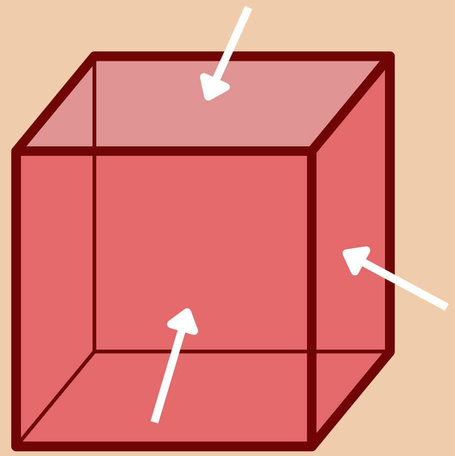


# Common 3D shapes



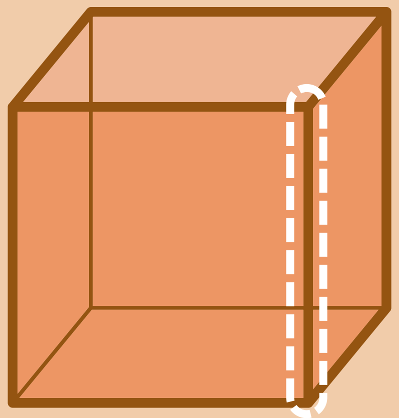
# Properties of 3D shape

## Faces



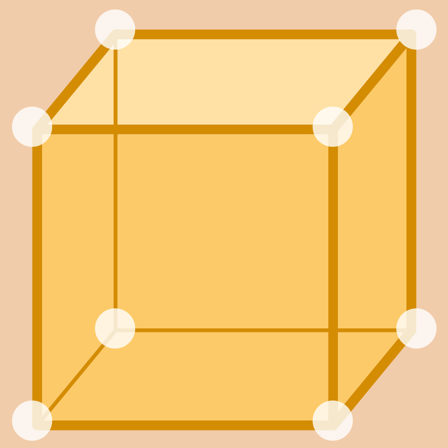
Faces are flat or curved surfaces on a 3D shape

## Edges



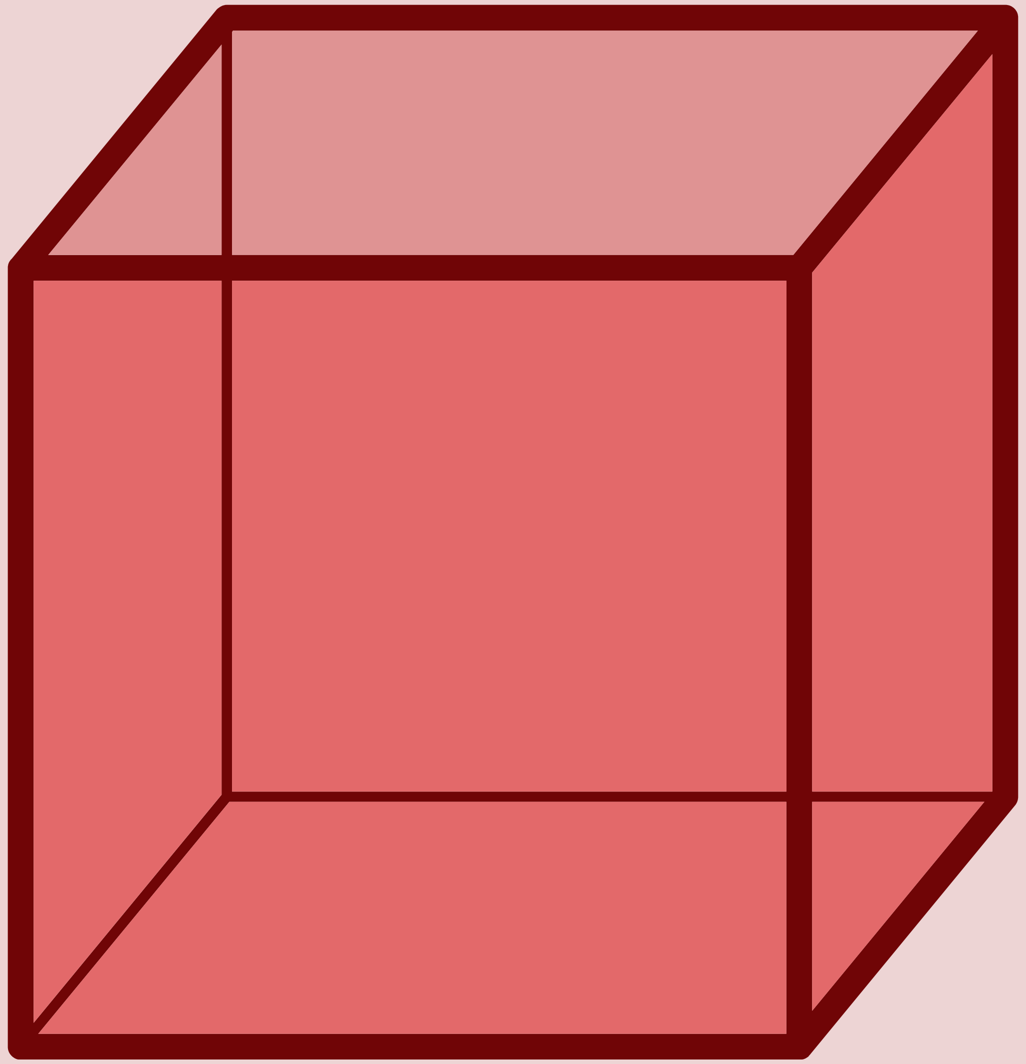
An edge is where two faces meet

## Vertices



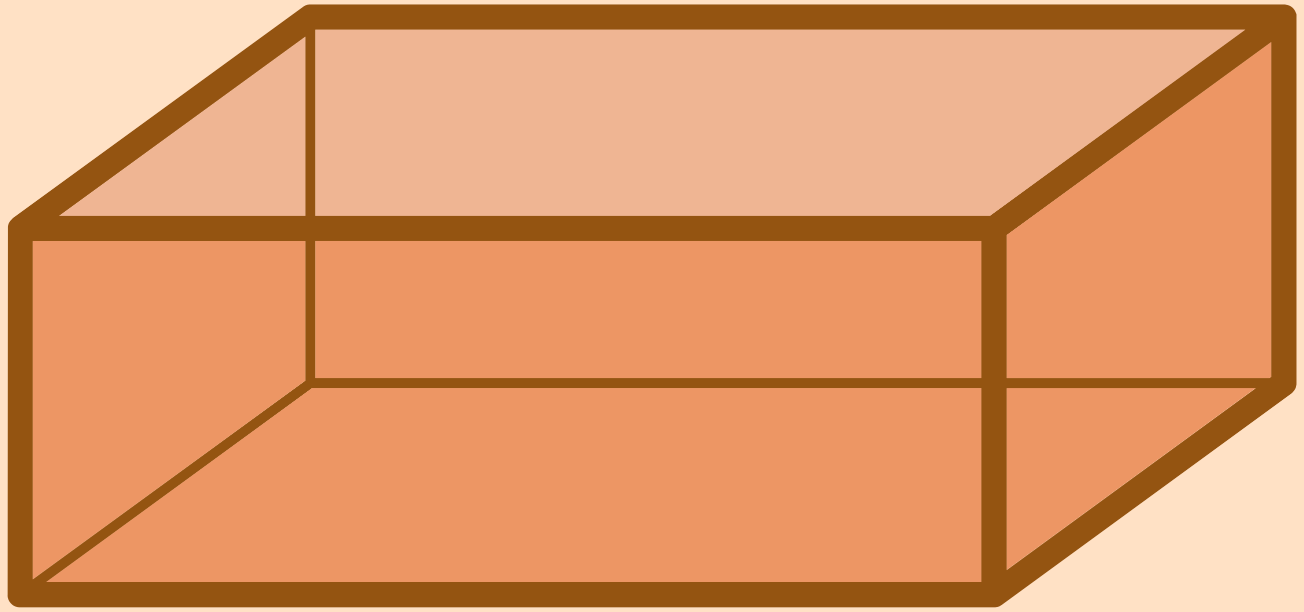
Vertices/corners are where two or more edges meet

# Cube



| | |
|---|---|
| Faces | 6 |
| Edges | 12 |
| Vertices | 8 |

# Cuboid

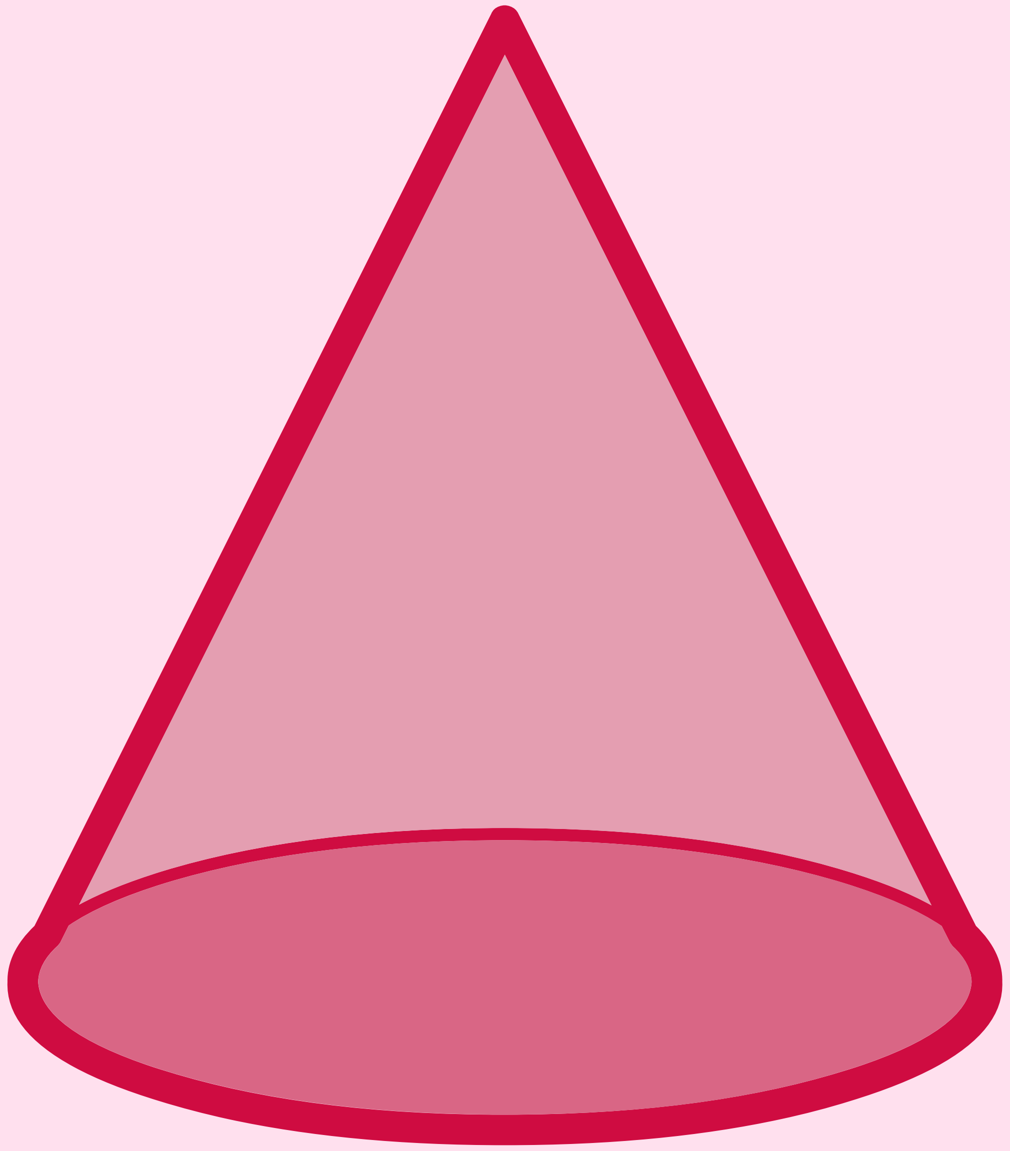


| | |
|---|---|
| Faces | 6 |
| Edges | 12 |
| Vertices | 8 |

# Sphere

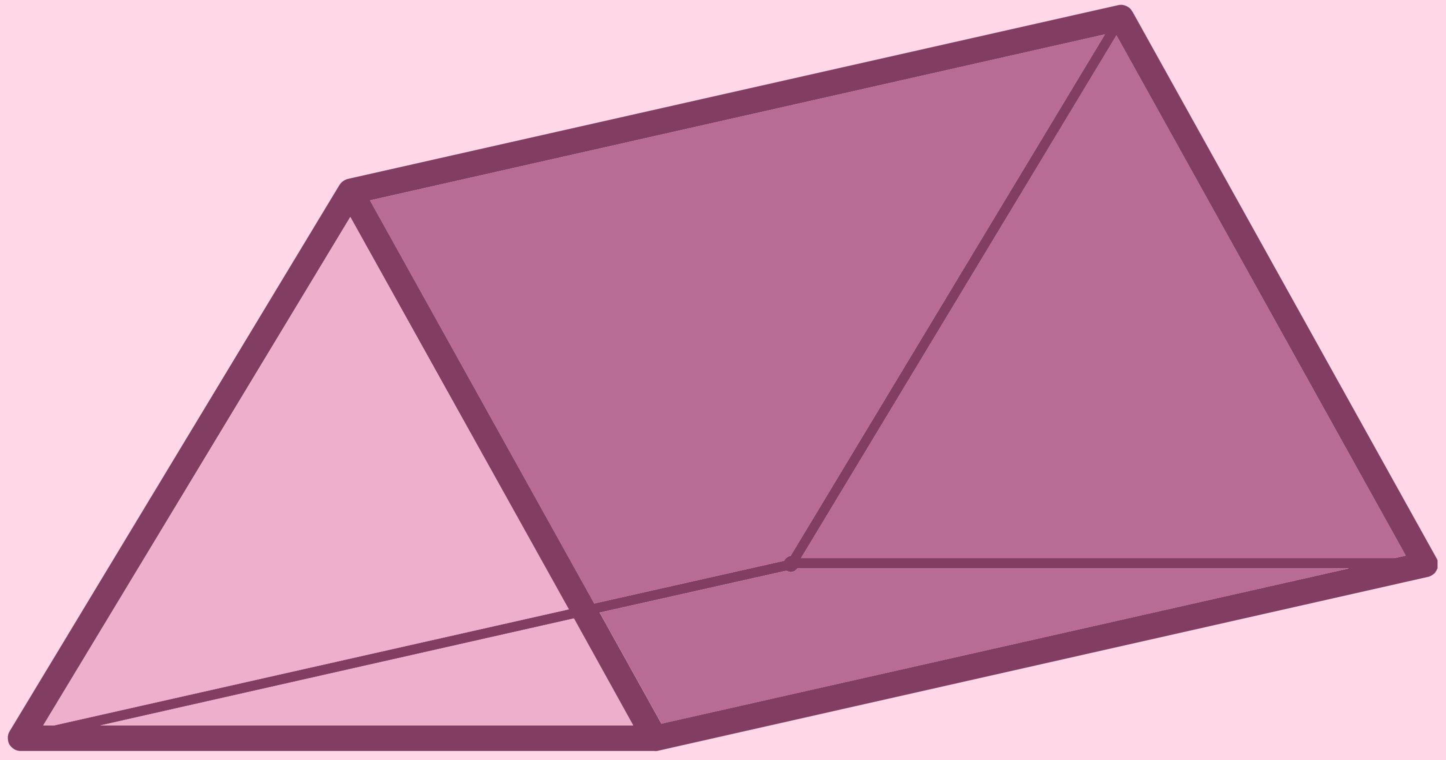
| | |
|---|---|
| Faces | 1 |
| Edges | 0 |
| Vertices | 0 |

# Cone



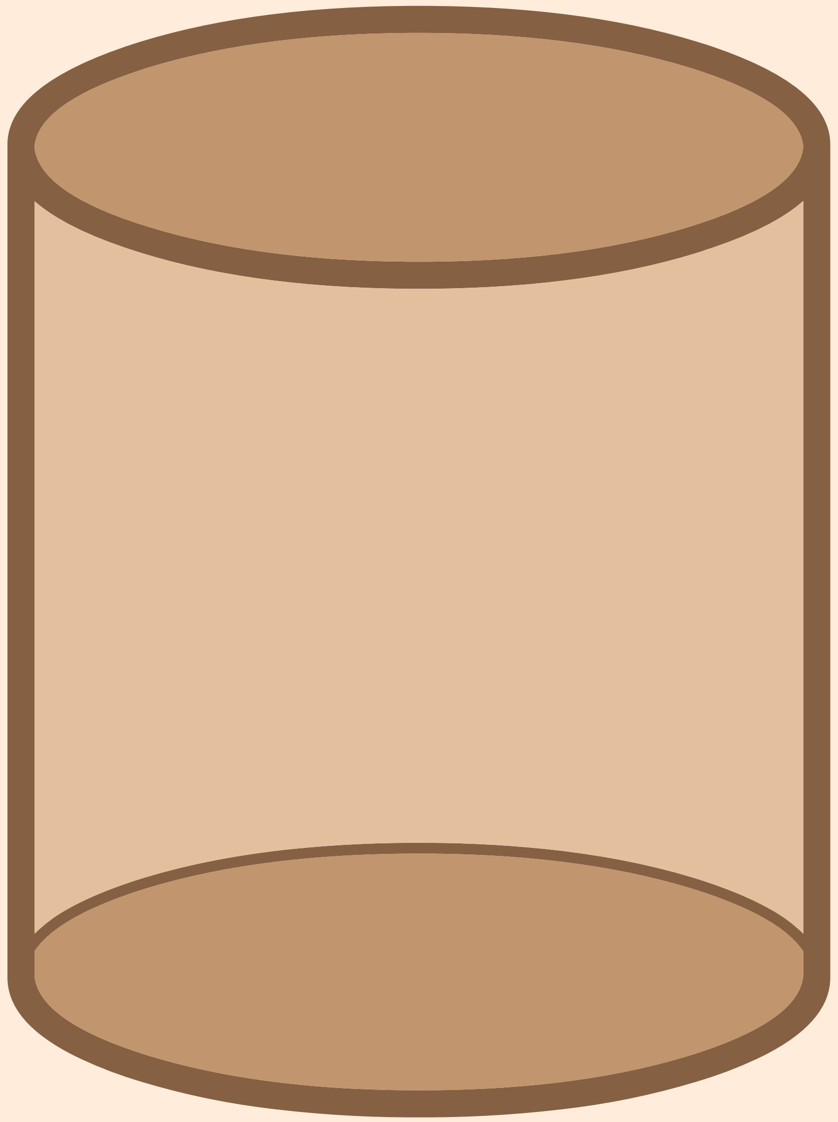
| | |
|---|---|
| Faces | 2 |
| Edges | 1 |
| Vertices | 1 |

# Triangular Prism



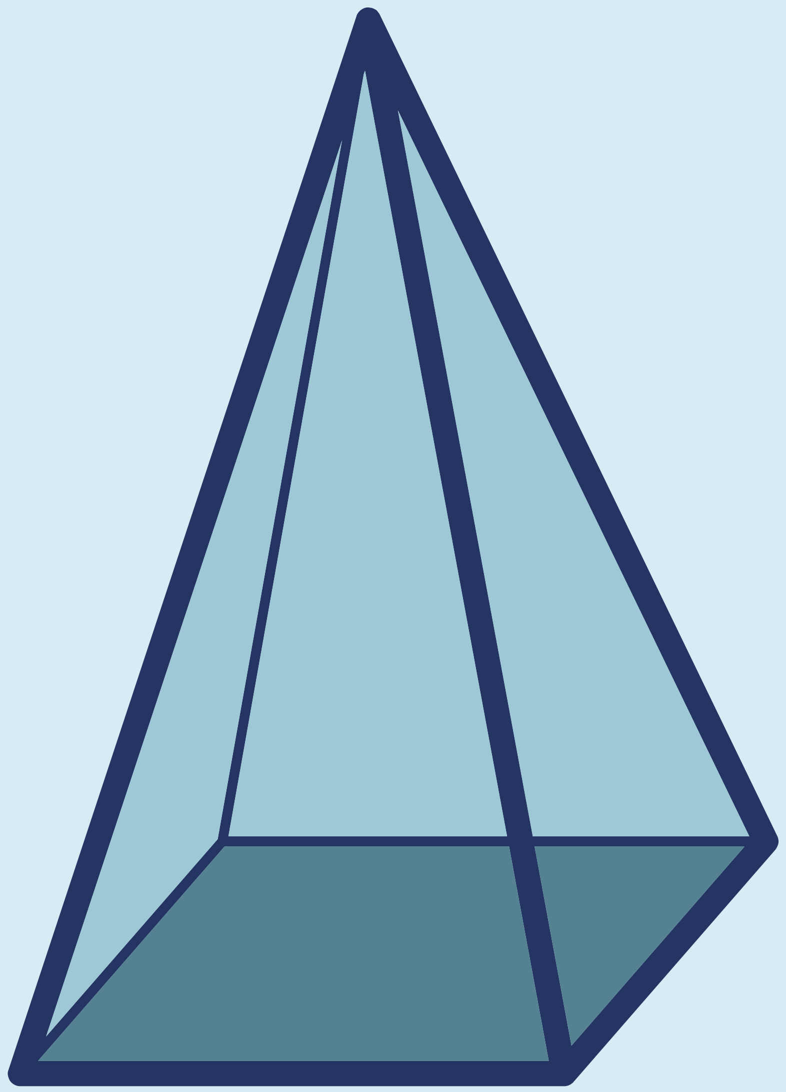
| | |
|---|---|
| Faces | 5 |
| Edges | 9 |
| Vertices | 6 |

# Cylinder



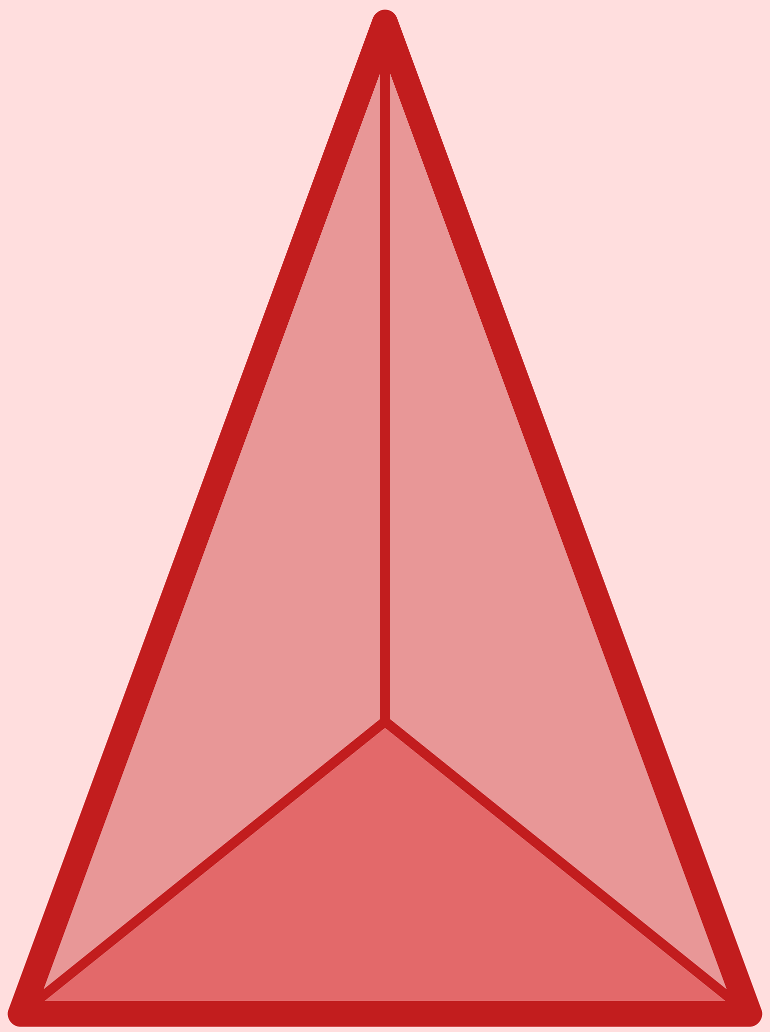
| | |
|---|---|
| Faces | 3 |
| Edges | 2 |
| Vertices | 0 |

# Square-based Pyramid



| | |
|---|---|
| Faces | 5 |
| Edges | 8 |
| Vertices | 5 |

# Tetrahedron



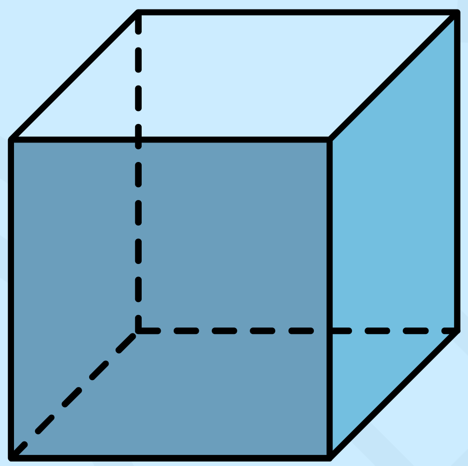
| | |
|---|---|
| Faces | 4 |
| Edges | 6 |
| Vertices | 4 |

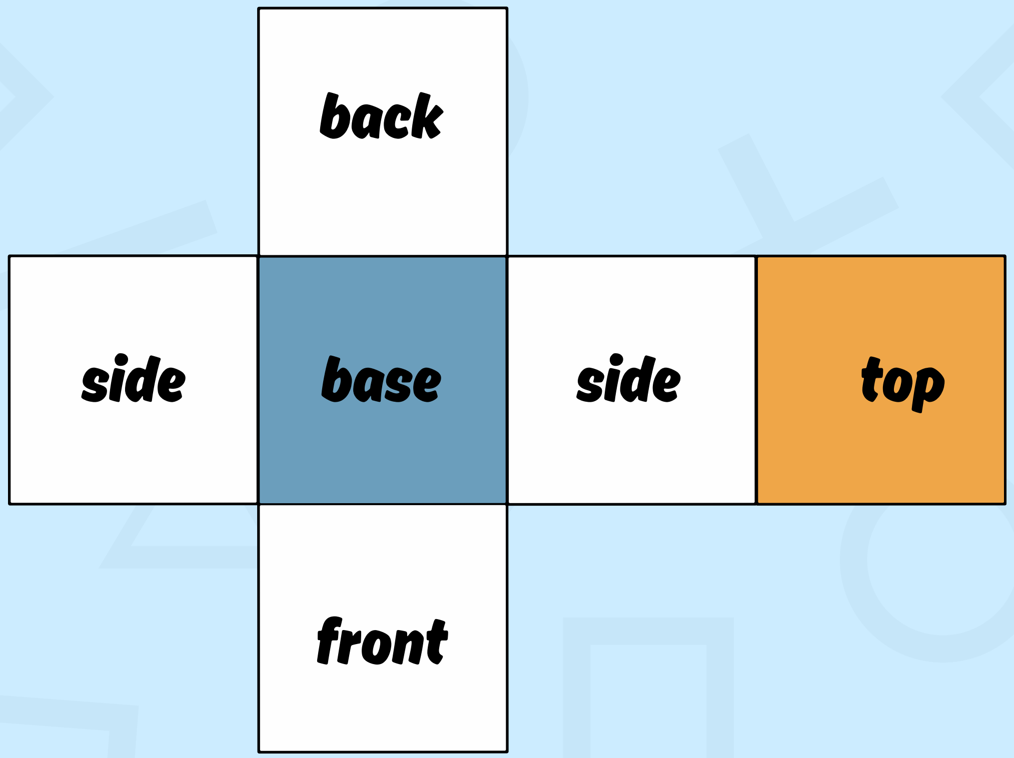
## What is a net?

A net is a three-dimensional shape that has been flattened out and opened up. It may be folded to form a 3D shape and indicates the number and type of faces a 3D shape contains.

# Net of a Cube

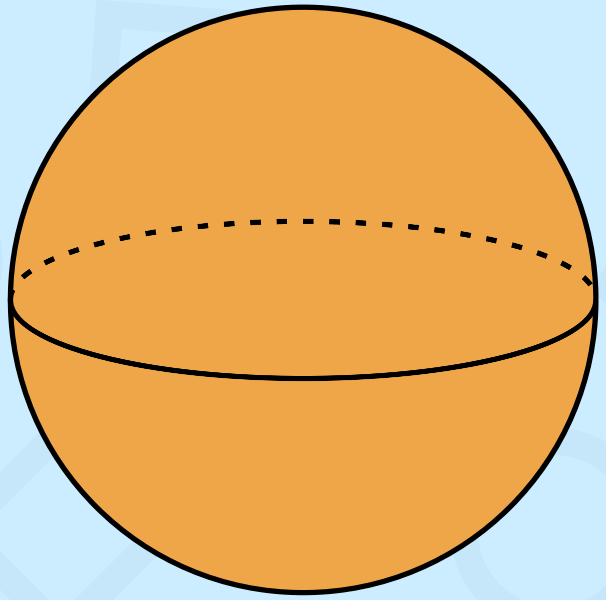
- A cube has 6 faces.
- A cube consists of 6 square faces that are connected together.
- When a cube is unfolded it will give 6 squares.
- All faces are congruent and perpendicular to each other.
- There are 11 nets in a cube.

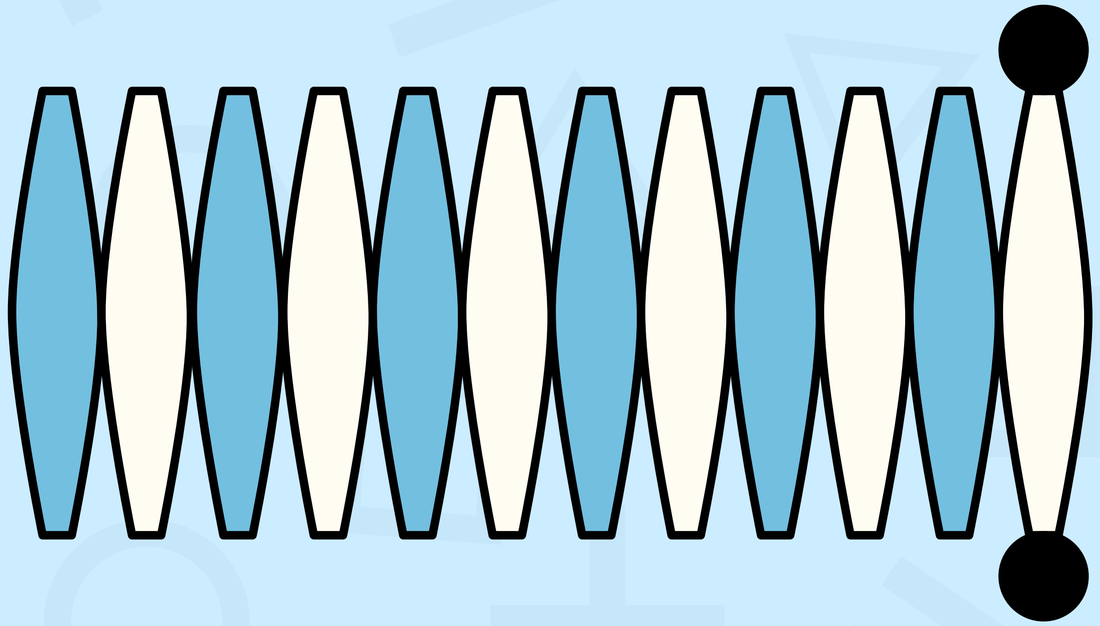




# Net of a Sphere

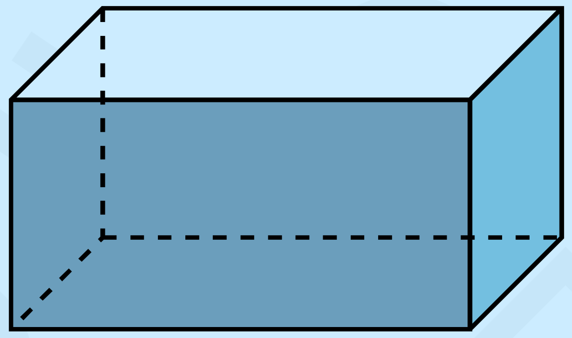
- A sphere is a perfectly symmetrical 3d shape that has no flat surface.
- It has all point equidistant from the center.
- When a sphere is unfolded, it will give a continuous curve.

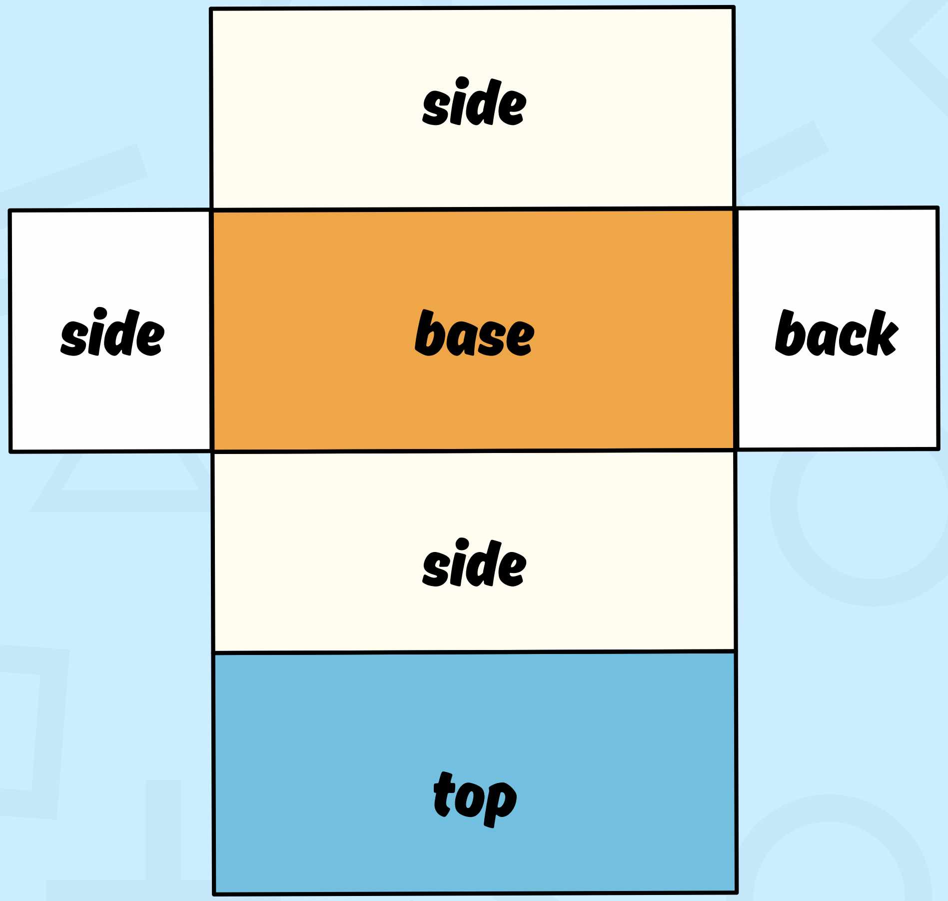




# Net of a Cuboid

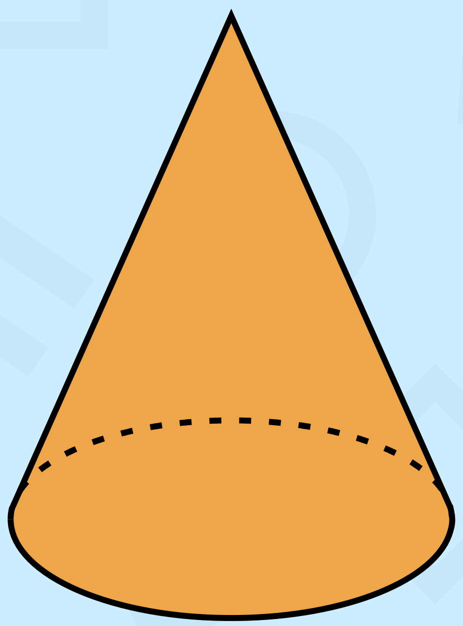
- A cuboid has 6 faces and all of these faces are rectangles
- If a cuboid is unfolded, it will give 6 rectangles.
- Opposite faces are congruent and parallel to each other.
- There are 54 different nets of the cuboid.

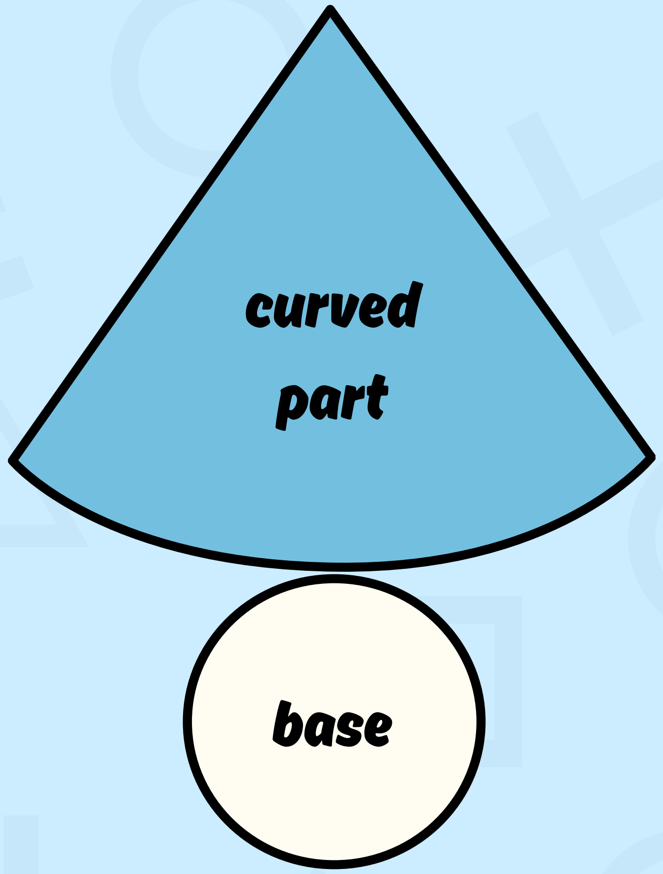




# Net of a Cone

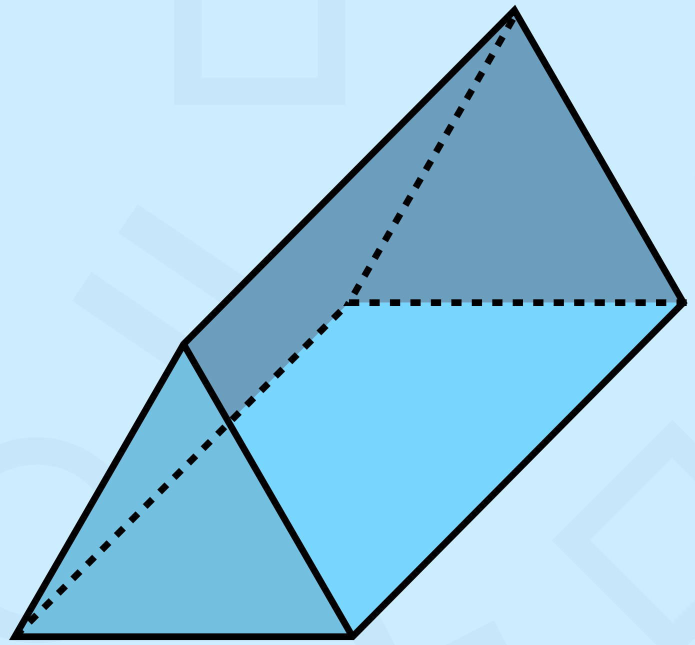
- The cone has only 2 faces.
- Out of the 2 faces, 1 is a curved surface and the other one is a flat face.
- A net of cones has 2 parts: One is the circle that represents the flat part. The second one represents the curved part.
- When it is unfolded, it will give 1 circle and 1 sector that tapers to a single point called apex.

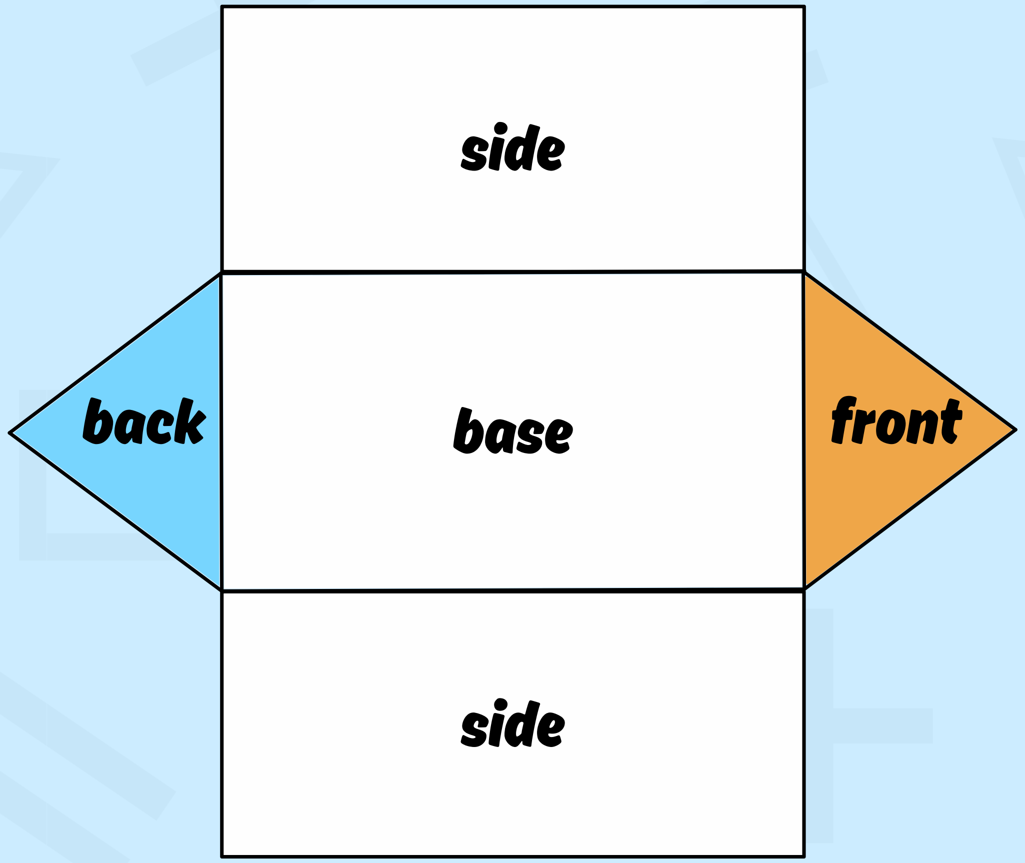




# Net of a Prism

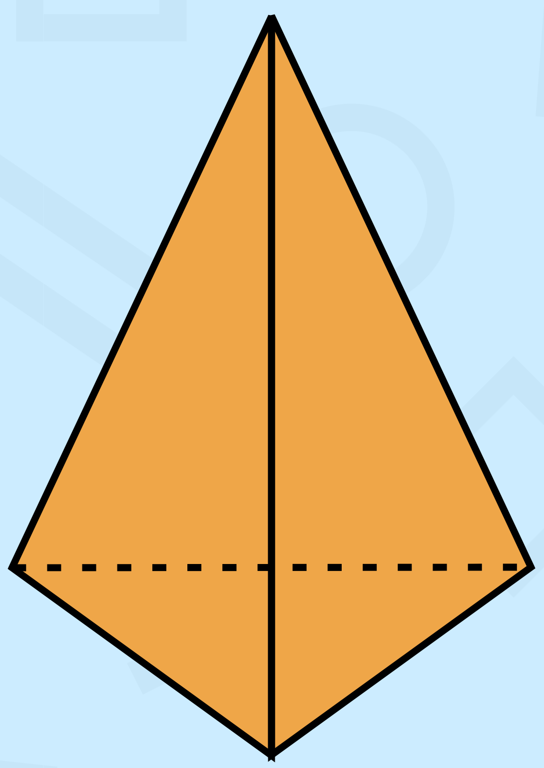
- A prism has 2 identical, congruent polygons that form from the top and bottom of the prism.
- The triangular prism has 2 triangular bases and 3 rectangular lateral faces connecting the corresponding edges of the triangles.

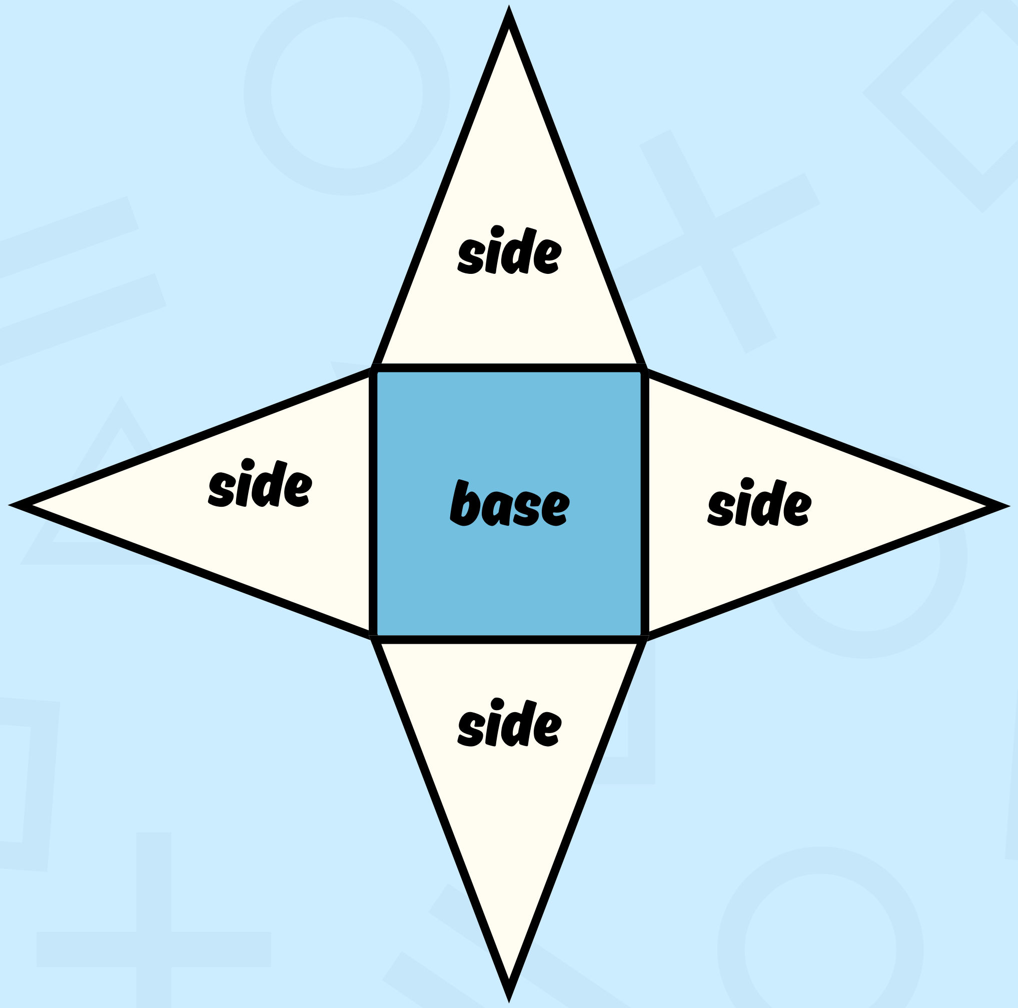




# Net of a Pyramid

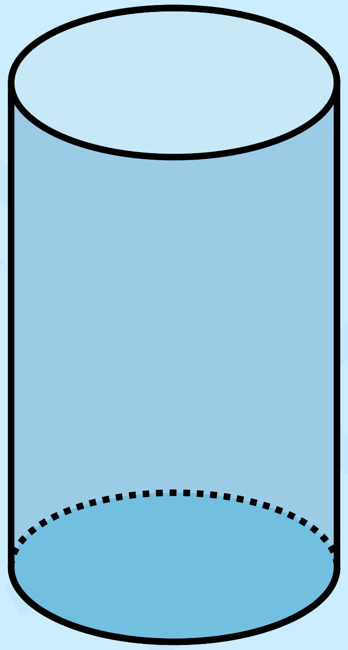
- A pyramid has 5 faces.
- These faces include 1 square and 4 triangular faces.
- When a pyramid is unfolded it will give 1 square and 4 triangles.

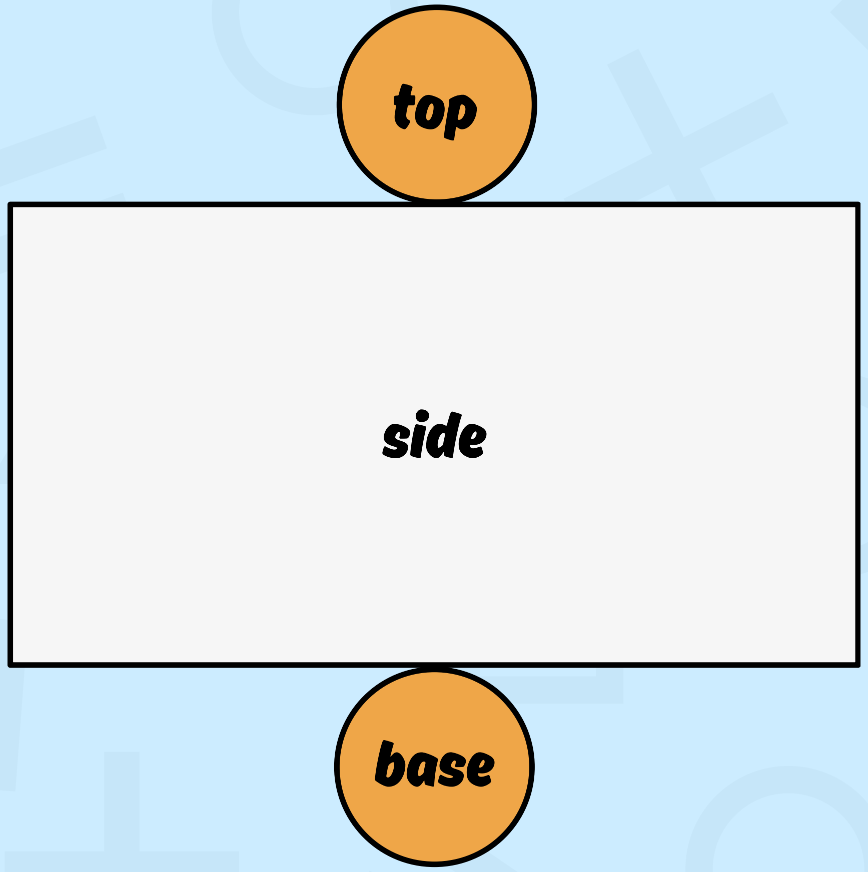




# Net of a Cylinder

- A cylinder is made up of 3 faces.
- Two are circular bases and one is a curved surface.
- Two of the faces are circular and one is rectangular.
- When a cylinder is unfolded, it will give 2 circles and 1 rectangle.





# REAL-LIFE APPLICATIONS OF GEOMETRY

- Architecture: Used in building designs and structures.
- Engineering: Helps in designing machines and structures.
- Navigation: GPS and maps use geometric concepts.
- Art and Design: Shapes and symmetry are key in creative fields.

# CONCLUSION

- Geometry is everywhere!
- It helps us understand and create the world around us.
- Whether in nature, technology, or design, geometry plays a crucial role in shaping our lives.

Grade 4

# See you next time!

Thank you!